Between Metros, Not Within:
The Racial Incidence of Identified Monetary Contractions on Local Employment

Abdallah Dalis*

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Abstract

Monetary contractions reduce employment, but whether the cost falls disproportionately on communities of color is contested and rarely estimated with clean identification. Using Jorda local projections across 2,884 U.S. counties from 2002 to 2019, I interact identified Bauer–Swanson monetary policy surprises with predetermined 2000 Census racial composition. Counties with higher Black population shares experience larger employment contractions after an identified tightening; the gradient is robust to controls for county size, income, and industry exposure, and income strengthens rather than attenuates it. The estimate is directionally stable but imprecise, as expected from a short high-frequency surprise series. A within-metropolitan decomposition is decisive: replacing national time effects with metropolitan-by-quarter effects reverses the sign. The gradient is a between-metropolitan phenomenon. Higher-Black metros contract more as whole labor markets, while within a metro, higher-Black counties do not. The racial incidence of monetary contractions operates across metropolitan areas, not within them, pointing to regional and industrial structure rather than a within-market race effect.

Keywords: monetary policy transmission; local labor markets; racial inequality; local projections; high-frequency identification.
JEL: E52, J15, R23, E24.

*DePaul University. Email: `dalisabdallah@gmail.com`. Replication code and data: `Dalis_Abdallah_Part3_Code.R`. This is the third installment of a research program on the local labor-market transmission of monetary policy. I thank Prof. Jin Man Lee for advising the program. All errors are my own.

Plain-Language Summary

When the Federal Reserve raises interest rates to slow the economy, some people lose their jobs. This paper asks a simple question: do places with more Black residents lose more of those jobs? To answer it credibly, I use only the *surprise* part of Fed decisions, the piece markets did not see coming, so that I am measuring the effect of policy rather than a coincidence.

Three things emerge. First, counties with larger Black populations do tend to lose more jobs after a surprise rate increase. The direction matches the concern, though the data is noisy enough that the result is not statistically airtight. Second, this is not simply a story about poorer, smaller, or differently-industried places: the pattern holds when those factors are held constant, and accounting for income actually makes it sharper.

Third, and most important, the disadvantage shows up *between cities*, not between neighbors. Heavily-Black metropolitan areas, taken as whole labor markets, contract more than other metros. But inside a single metro, a county with a larger Black share is not hit harder than a whiter county next door. So the cost is tied to which metropolitan economy a community belongs to, with its particular industries and regional exposure, rather than to race operating block by block.

The honest limit is precision. With about eighteen years of these surprises, the pattern is clear in direction and survives every reasonable objection, but it cannot be pinned to an exact number. The contribution is a clean, well-identified design and a robust structural finding about *where* the disadvantage lives.

1 INTRODUCTION

Monetary policy is not distributionally neutral. A growing literature documents that Black employment is markedly more cyclically sensitive than white employment (Cajner et al., 2017), and that episodes of monetary tightening widen racial gaps in labor market outcomes (Carpenter and Rodgers, 2004; Seguino and Heintz, 2012). A parallel literature shows that the transmission of policy is geographically uneven, varying with regional industrial composition (Carlino and DeFina, 1998). Together these strands raise a natural question about *incidence*: when contractionary policy destroys jobs, who bears the loss?

Two obstacles have kept the question hard to answer cleanly. First, distributional estimates typically use the realized federal funds rate or other aggregate series, which over the 2002–2019 window are contaminated by endogeneity and by the information effect, the tendency of announced policy to bundle the central bank's private read on the economy. The earlier installments of this program show that, in this window, the endogenous funds rate even delivers a perverse positive employment response that the identified surprise corrects. Second, a county-level racial gradient can reflect race or any of the things race is correlated with: county size, income, industry mix, and region. Distinguishing these requires both an identified shock and a research design that isolates the comparison.

This paper supplies both. I hold fixed the identified design of Part 2, a Jorda local projection driven by the Bauer–Swanson high-frequency monetary policy surprise (Bauer and Swanson, 2023; Jorda, 2005), and swap the heterogeneity dimension from industry exposure to predetermined 2000 Census racial composition. I then build a control ladder (size, income, industry exposure) and, decisively, a within-metropolitan comparison that holds the metro-quarter fixed.

Three findings follow. (i) The surprise-by-Black-share gradient is negative and deepens with the horizon, consistent with disproportionate incidence, but is imprecise. (ii) It survives the control ladder; income sharpens rather than absorbs it. (iii) It is a between-metropolitan phenomenon: within-metro identification reverses the sign. The disadvantage is attached to which metropolitan economy a community sits in, not to its racial composition relative to neighbors. The remainder of the paper presents the data (Section 2), the estimator and identification (Section 3), the results figure by figure (Section 4), and a discussion of mechanisms, caveats, and what the between-metro location of the effect implies (Sections 5–6).

2 LITERATURE REVIEW

This paper sits at the intersection of three literatures: the high-frequency identification of monetary policy shocks, the regional and local transmission of those shocks, and their distributional incidence across racial groups.

2.1 Identifying monetary policy shocks

A core difficulty in measuring the effects of monetary policy is that the policy rate responds to the economy, so realized rate changes confound policy with the conditions that provoked it. Jorda (2005) introduced local projections as a flexible alternative to vector autoregressions for tracing dynamic responses, estimating a separate regression at each horizon and avoiding the recursive extrapolation that makes VAR impulse responses fragile; this is the estimator I adopt. High-frequency identification narrows the remaining endogeneity by measuring the surprise component of policy in a tight window around announcements, but Nakamura and Steinsson (2018) show that these surprises are themselves contaminated by an information effect: a tightening surprise can raise private output forecasts, as if the announcement reveals the central bank's favorable read on the economy, biasing estimated effects toward zero or the wrong sign. Bauer and Swanson (2023) offer an alternative explanation, arguing the apparent information effect reflects the central bank and the public responding to the same publicly available news, and construct an orthogonalized surprise series that purges this predictable component. I use their orthogonalized surprise as the identified shock, which is what allowed Part 2 of this program to overturn the perverse positive employment response that the endogenous funds rate produces over the 2002–2019 window.

2.2 Regional and local transmission

Monetary policy does not land evenly across places. Carlino and DeFina (1998) document that the employment response to policy varies systematically across U.S. regions, with the differences traceable to regional variation in industry mix and the interest sensitivity of local production. This motivates a place-based design but also warns that any cross-sectional gradient may reflect industrial composition rather than the characteristic of interest. The natural design for such gradients is the shift-share or Bartik approach (Bartik, 1991), in which a local exposure share is interacted with an aggregate shock; Goldsmith-Pinkham et al. (2020) clarify that identification in these designs rests on the exogeneity of the predetermined shares. Parts 1 and 2 of this program applied exactly this logic with industry-exposure shares and found that the exposure gradient did not survive controlling for county size. Part 3 keeps the identified-shock-times-predetermined-share structure but replaces industry exposure with racial composition.

2.3 Distributional incidence by race

A distinct literature asks who bears the employment cost of contractions. Carpenter and Rodgers (2004), using recursive VARs and distributed-lag models on aggregate series, find that the employment of minorities, and of African Americans in particular, is more sensitive to contractionary policy than that of whites, operating mainly through higher unemployment rather than reduced participation. Seguino and Heintz (2012), using U.S. state-level panel data from 1979 to 2008, find that the unemployment cost of monetary tightening falls most heavily on Black women and Black men, that the effect is not race-neutral, and, most relevant here, that the relative cost borne by subordinate groups *varies with the Black share of the population*. Cajner et al. (2017) provide the background regularity: Black unemployment is both higher and substantially more cyclical than white unemployment, driven largely by a higher risk of job loss, while the Hispanic-white gap is smaller and mostly explained by educational composition, a pattern that anticipates the null Hispanic gradient I find.

2.4 The gap this paper fills

Prior distributional work is built almost entirely on national time-series or state-level panels, and on the realized policy rate or unidentified surprises. Two gaps follow. First, the distributional question has not been asked with a cleanly identified shock at a fine geographic scale, leaving open whether the documented racial gradient reflects policy or the endogeneity and information-effect problems that Part 2 shows are first-order in this window. Second, and more fundamentally, the existing place-based estimates do not separate *between*-metropolitan from *within*-metropolitan variation. Seguino and Heintz (2012) show the cost varies with the Black population share, but a state-level design cannot tell whether that gradient operates across metropolitan labor markets or between adjacent counties inside one. This paper supplies both: an identified surprise interacted with predetermined county racial composition, a control ladder that nets out size, income, and industry exposure, and a within-metro comparison that locates where the gradient actually lives.

3 DATA

The estimation sample is a panel of 2,884 U.S. counties observed quarterly from 2002Q1 through 2019Q4 (72 quarters), inherited intact from Part 2.

Employment. County employment is the QCEW quarterly establishment count (`month3_emplvl`); the outcome at horizon h is the log-employment change $100\,(\ell_{i,t+h} - \ell_{i,t-1})$.

Identified monetary shock. The Bauer–Swanson orthogonalized high-frequency monetary policy surprise (Bauer and Swanson, 2023), summed to quarterly frequency. The orthogonalization purges the surprise of the predictable macro-news component that drives the Fed information effect.

Predetermined heterogeneity dimensions. County racial composition is the predetermined 2000 Census share Black-alone and Hispanic (April 1, 2000 population base), so the shares cannot respond to anything in the 2002–2019 sample. Predetermined controls: log 2000 population (size); log 2000 median household income (Census 2000 SF3, table P053001); and the Part 1–2 industry-exposure share (2002 CBP construction-plus-manufacturing employment share). All dimensions are standardized, so coefficients are read per one-standard-deviation of the share.

Metropolitan grouping. Counties are mapped to Core-Based Statistical Areas via the Census 2020 delineation. The within-metro analysis retains counties in a CBSA with at least two sample counties (1,110 counties across 300 CBSAs), since metro-by-quarter effects require within-metro variation.

4 METHODOLOGY

The estimator is a Jorda local projection (Jorda, 2005). For each horizon $h = 0, \ldots, 12$ quarters,

$$100\left(\ell_{i,t+h} - \ell_{i,t-1}\right) \;=\; \alpha_i^h + \delta_t^h \;+\; \theta_h\left(s_t \times d_i\right) \;+\; \mathbf{x}_i'\gamma_h \;+\; \varepsilon_{i,t+h}, \tag{1}$$

where s_t is the identified surprise, d_i a standardized predetermined county characteristic (Black share in the headline specification), α_i^h county fixed effects, and δ_t^h time fixed effects. Because s_t is national, the time effects absorb its main effect; θ_h is identified from the interaction of the shock with cross-county differences in d_i. A negative θ_h means higher-share counties contract more after an identified tightening.

Control ladder. I add $s_t \times \log \mathrm{pop}_i$, $s_t \times \log \mathrm{inc}_i$, and $s_t \times \mathrm{exposure}_i$ sequentially, so the racial gradient is read net of the county-size confound that absorbed the Part 2 industry gradient, of income, and of the industry channel.

Within-metro identification. The key decomposition replaces the national time effect δ_t^h with a metro-by-quarter effect $\delta_{c(i),t}^h$, comparing higher- and lower-Black counties *inside the same metro-quarter*. County and metro-by-quarter effects are non-nested and absorbed by `fixest::feols`.

Inference. Standard errors are clustered by quarter in the baseline. As a robustness check I report Driscoll–Kraay standard errors (Driscoll and Kraay, 1998), which additionally allow serial correlation across quarters. The surprise series spans 72 quarters, so estimates are expected to be imprecise; I report this rather than select favorable horizons.

Reproducibility. All results are produced by `Dalis_Abdallah_Part3_Code.R` (Sections 1–7) and were verified on the author's machine.

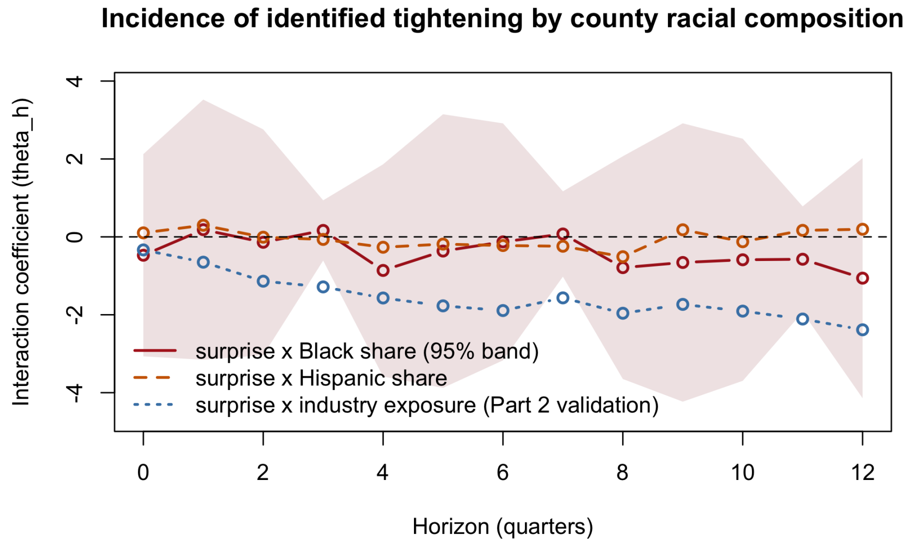

Figure 1: Interaction coefficient θ_h from Eq. (1): identified surprise $\times$ standardized Black share (with 95% band), Hispanic share, and industry exposure (Part 2 validation). County and quarter fixed effects; quarter-clustered SE.

5 RESULTS

5.1 The baseline gradient and a validation check

Figure 1 plots θ_h for the surprise interacted with Black share, Hispanic share, and (as a validation against Part 2) industry exposure. The exposure line reproduces Part 2: negative and monotonically deepening to -2.39 by h=12, confirming the pipeline. The Black-share gradient is negative at ten of thirteen horizons, deepening to roughly -1.06 by h=12, but statistically insignificant throughout ($|t| < 0.9$). Hispanic share is a null at every horizon. The direction is consistent with disproportionate incidence; the precision is not.

5.2 The control ladder: income sharpens, not absorbs

Table 1 and Figure 2 report the Black coefficient as controls are added. The gradient survives netting out county size, the confound that eliminated the Part 2 industry gradient. Adding 2000 median household income does not absorb it; it *sharpens* it at the long horizons (from -1.06 to -1.19 at h=12, with $|t|$ approaching 1.0 at h=11), because poorer and higher-Black counties partly offset, so holding income fixed widens the racial gap. Adding industry exposure leaves it essentially unchanged. The gradient is therefore not size, income, or industry mix in disguise; it remains imprecise.

5.3 Between metros, not within

Figure 3 and Table 2 report the decomposition that reframes the paper. On the identical metro-county sample, replacing the national time effect with a metro-by-quarter effect flips the sign

Table 1: Black-share coefficient θ_h across the control ladder (national time FE, full sample $n = 2{,}884$ counties). Selected horizons; t-statistics in parentheses, quarter-clustered.

Horizon	Black alone	+ logpop	+ logpop+inc	+ logpop+inc+exp
h=0	−0.47 (−0.36)	−0.46 (−0.40)	−0.46 (−0.56)	−0.46 (−0.56)
h=4	−0.86 (−0.62)	−0.66 (−0.55)	−0.70 (−0.73)	−0.70 (−0.72)
h=8	−0.79 (−0.54)	−0.54 (−0.43)	−0.75 (−0.69)	−0.75 (−0.68)
h=11	−0.57 (−0.83)	−0.38 (−0.79)	−0.93 (−1.02)	−0.92 (−1.03)
h=12	−1.06 (−0.67)	−0.77 (−0.58)	−1.19 (−0.90)	−1.18 (−0.90)

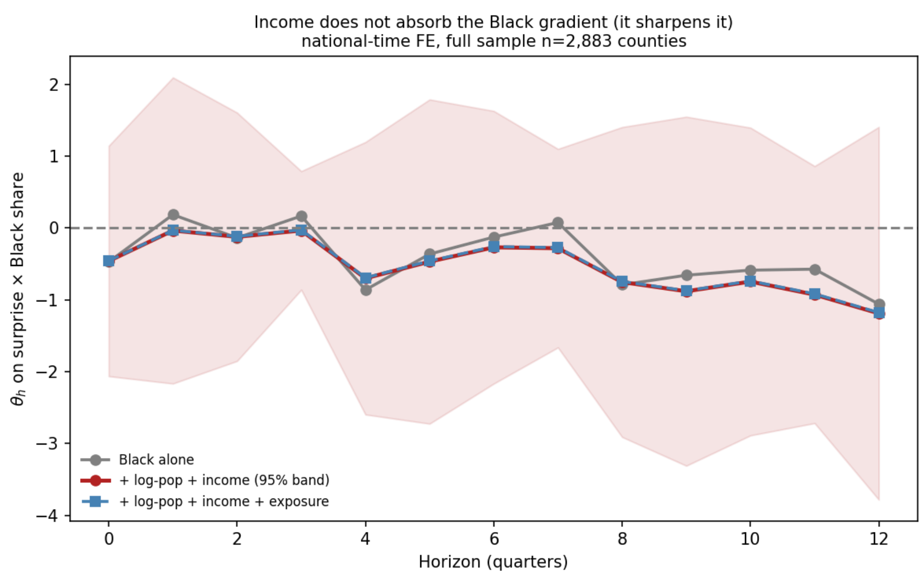

Figure 2: The Black-share gradient under successive controls. Income does not absorb the gradient; it sharpens it at long horizons. National time FE, full sample.

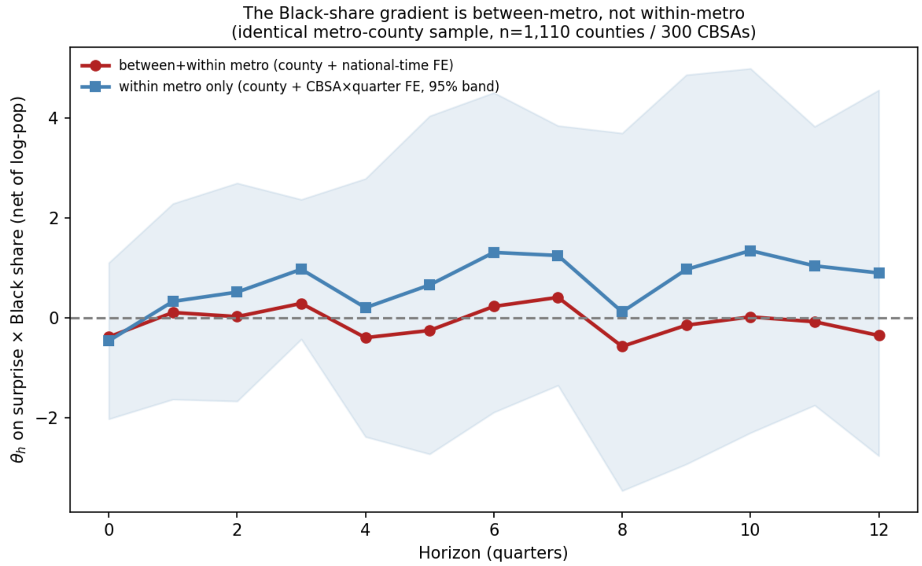

Figure 3: Black-share gradient, identical metro-county sample ($n = 1{,}110$ counties, 300 CBSAs): between-plus-within (county + national-time FE) vs within-metro (county + metro×quarter FE, 95% band).

Table 2: Between- vs within-metro Black-share coefficient θ_h (metro sample, net of log population). Selected horizons; quarter-clustered SE in parentheses.

Horizon	Between + within (national-time FE)	Within metro (metro×quarter FE)
h=0	−0.39 (1.02)	−0.46 (0.80)
h=4	−0.40 (1.13)	0.20 (1.32)
h=8	−0.58 (1.14)	0.11 (1.83)
h=12	−0.36 (1.17)	0.89 (1.87)

of the Black gradient from negative to positive (at h=12, from −0.36 between-and-within to +0.89 within-metro). Because only the fixed-effect structure changes, this is not a sample artifact. The negative gradient lives in the *between-metropolitan* variation: metros with higher Black population shares contract more after identified tightening. Inside a given metro, higher-Black counties do not. All estimates remain imprecise.

5.4 Robustness

Two checks complete the picture. First, entering Black and Hispanic shares jointly leaves the Black gradient essentially unchanged (−1.17 at h=12) while Hispanic remains null, so the gradient is not Hispanic composition in disguise. Second, Driscoll–Kraay standard errors do not weaken the result; at the long horizons they are if anything tighter than the clustered errors ($|t| \approx 1.5$ at h=11). The imprecision is a property of the 72-quarter surprise series, not of the inference method or the controls.

6 DISCUSSION

Review. The headline is a direction that is robust and a magnitude that survives controls, paired with precision that never crosses conventional thresholds, and a decomposition that locates the entire negative gradient between metropolitan areas rather than within them.

Interpretation. A between-metro racial gradient that does not survive within-metro comparison points away from a within-labor-market race mechanism and toward the characteristics that distinguish higher-Black metros: industrial composition, regional exposure, and local policy regimes. This rhymes with the program's arc. Part 2 was "identification, not aggregation"; Part 3 is "between, not within." It is consistent with a literature in which Black employment is more cyclically exposed (Cajner et al., 2017) and tightening widens racial gaps (Carpenter and Rodgers, 2004; Seguino and Heintz, 2012), while clarifying that, in this sample, the place-based version of that exposure is a cross-metro phenomenon.

Caveats and risks.

- **Power.** Seventy-two quarters of an identified surprise leave every estimate imprecise. The contribution is a clean, well-identified design and a robust *pattern*, not a precise point estimate.
- **Place vs people.** County composition proxies who bears the cost; place-level incidence is not worker-level incidence.
- **Information effect.** Residual Nakamura–Steinsson information-effect contamination of the surprise series (Nakamura and Steinsson, 2018) carries over from Part 2 and is the deepest unresolved threat; it is discussed, not claimed solved.

7 CONCLUSION

Identified monetary contractions fall harder on metropolitan areas with larger Black populations, taken as whole labor markets, but not on higher-Black counties relative to whiter counties within the same metro. The estimate is directionally robust, survives controls for size, income, and industry as well as joint entry of Hispanic share and a more demanding standard-error treatment, and is honestly underpowered given the length of the surprise series.

Two implications follow. For research, distributional analysis of monetary policy should condition on the metropolitan scale, since aggregating or pooling across metros can attribute to race what is really cross-metro structure, while a naive within-metro look can miss the disadvantage entirely. For policy, the lever that matters is the differential exposure of metropolitan economies, their industries and regions, rather than a county-level racial channel. The most promising next steps are a longer or extended surprise series to lift power, worker-level outcomes to move from place-based to person-based incidence, and a decomposition of the between-metro gradient into its industrial and regional components.

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